

Chapter 15

Summary of Curve Sketching

§ Section: 4.5

15.1 Summary of Curve Sketching

Guidelines for Sketching a Curve:

1. Domain: It's often useful to start by determining the domain of f .
2. Intercepts: The y -intercept is $f(0)$. To find x -intercepts, we set $y = f(x) = 0$ and solve for x .

$$f(-x) = f(x) \quad f(-x) = -f(x)$$
3. Symmetry: Is $f(x)$ even, odd, or periodic?
4. Asymptotes: Find all **Horizontal**, **Vertical**, or **Slant Asymptotes**.
5. Intervals of Increase or Decrease: Compute $f'(x)$ and use the I/D Test.
6. Local Maximum and Minimum Values.
7. Concavity and Points of Inflection: Compute $f''(x)$ and use the Concavity Test.
8. Sketch the Curve.

Definition:

If $\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$ or $\lim_{x \rightarrow -\infty} [f(x) - (mx + b)] = 0$ where $m \neq 0$, then the line $y = mx + b$ is called a **slant asymptote of the curve** $y = f(x)$.

Finding Slant Asymptotes:

1. Compute $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$. Denote this limit by m .
2. Compute $\lim_{x \rightarrow \infty} (f(x) - mx)$. Denote this limit by b .
3. If both limits m, b exist and $m \neq 0$, then $y = mx + b$ is a slant asymptote of $y = f(x)$ as $x \rightarrow \infty$.
4. Repeat Steps 1, 2, 3 for $x \rightarrow -\infty$ and find the slant asymptote in negative infinity.

Example 1.1

Find asymptotes of $y = f(x) = \frac{x^3}{x^2 - 2x}$.

Sol:
$$\begin{array}{r} x+2 \\ x^2-2x \overline{) x^3} \\ \underline{x^3-2x^2} \\ 2x^2 \\ \underline{2x^2-4x} \\ 4x \end{array} \quad f(x) = (x+2) + \frac{4x}{x^2-2x}$$

$$\lim_{x \rightarrow \pm\infty} f(x) - (x+2) = \lim_{x \rightarrow \pm\infty} \frac{4x}{x^2-2x} = 0$$

$\Rightarrow y = x+2$ is a S.A. of $y = f(x)$.

Example 1.2

Find slant asymptotes of $y = f(x) = \sqrt{x^2 + 6x} - 2x$.

Sol: ① $m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{\sqrt{x^2+6x}}{x} - 2 = \lim_{x \rightarrow \infty} \frac{x \sqrt{1+\frac{6}{x}}}{x} - 2 = -1.$

② $b = \lim_{x \rightarrow \infty} f(x) - (-x) = \lim_{x \rightarrow \infty} \sqrt{x^2+6x} - x = \lim_{x \rightarrow \infty} \frac{6x}{\sqrt{x^2+6x} + x} = \lim_{x \rightarrow \infty} \frac{6x}{x \sqrt{1+\frac{6}{x}} + x} = 3.$

$y = -x+3$ is a S.A. of $y = f(x)$.

③ $m = \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+6x}}{x} - 2 = \lim_{x \rightarrow -\infty} \frac{-x \sqrt{1+\frac{6}{x}}}{x} - 2 = -3$

④ $b = \lim_{x \rightarrow -\infty} f(x) - (-3x) = \lim_{x \rightarrow -\infty} \sqrt{x^2+6x} + x = \lim_{x \rightarrow -\infty} \frac{6x}{\sqrt{x^2+6x} - x} = -3$

$y = -3x-3$ is a S.A. of $y = f(x)$

Example 1.3

Find all asymptotes of $y = f(x) = \frac{2x}{1 - e^x}$.

$$\text{sol: } m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{2}{1 - e^x} = 0 \quad \Rightarrow y = 0$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2x}{1 - e^x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{2}{-e^x} = 0. \quad \text{is a H.A.}$$

$$m = \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} \frac{2}{1 - e^x} = 2 \quad \Rightarrow y = 2x \text{ is a S.A.}$$

$$b = \lim_{x \rightarrow -\infty} f(x) - 2x = \lim_{x \rightarrow -\infty} \frac{2x}{1 - e^x} - 2x = \lim_{x \rightarrow -\infty} \frac{2x e^x}{1 - e^x} = \lim_{x \rightarrow -\infty} \left(\frac{2}{1 - e^x} \right) \left(\frac{x}{e^{-x}} \right) = 2 \cdot 0 = 0$$

Example 1.4

Find all asymptotes of $y = f(x) = x \arctan x$. Sketch the graph $y = x \arctan x$.

$$\textcircled{1} m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}.$$

$$\textcircled{2} b = \lim_{x \rightarrow \infty} f(x) - \frac{\pi}{2}x = \lim_{x \rightarrow \infty} x \left(\tan^{-1} x - \frac{\pi}{2} \right) = \lim_{x \rightarrow \infty} \frac{\tan^{-1} x - \frac{\pi}{2}}{\frac{1}{x}} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{1+x^2}}{-\frac{1}{x^2}} = -1.$$

$$\Rightarrow y = \frac{\pi}{2}x - 1 \text{ is a S.A.}$$

$$\textcircled{1} m = \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}.$$

$$\textcircled{2} b = \lim_{x \rightarrow -\infty} f(x) + \frac{\pi}{2}x = \lim_{x \rightarrow -\infty} \frac{\tan^{-1} x + \frac{\pi}{2}}{\frac{1}{x}} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow -\infty} \frac{\frac{1}{1+x^2}}{-\frac{1}{x^2}} = -1.$$

$$\Rightarrow y = -\frac{\pi}{2}x - 1 \text{ is a S.A.}$$

$$f'(x) = \tan^{-1} x + \frac{x}{1+x^2} \Rightarrow \begin{array}{l} f'(x) > 0 \text{ for } x \in (0, \infty) \\ f'(x) < 0 \text{ for } x \in (-\infty, 0) \end{array}$$

$$f''(x) = \frac{1}{1+x^2} + \frac{(1+x^2) - x \cdot 2x}{(1+x^2)^2} = \frac{(1+x^2) + (1-x^2)}{(1+x^2)^2} = \frac{2}{(1+x^2)^2} > 0$$